Interplay between chemical order and magnetic properties in $L1_0$ FeNi phase (tetrataenite): A First-Principles study

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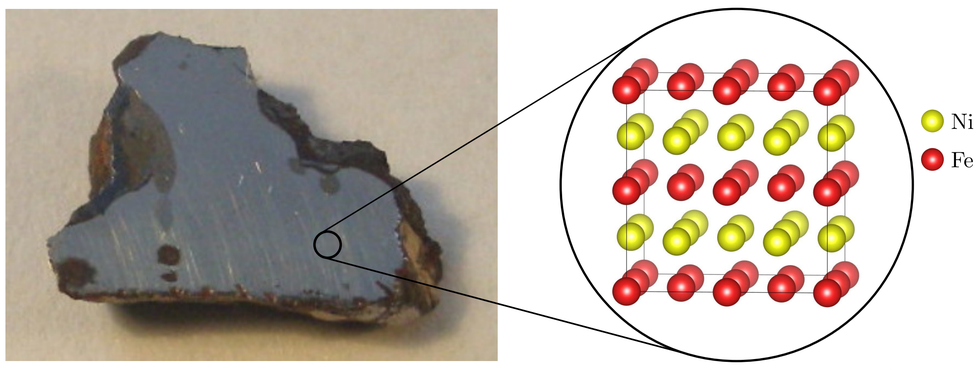
Introduction

Motivation

- High-cost of rare-earth-based permanent magnets has sparked great interest to explore alternative magnetic materials that are free from rare-earth elements
- Chemically-ordered $L1_0Fe_{50}Ni_{50}$phase (tetrataenite) found in iron meteorites is a promising magnetic material with large uniaxial magnetic anisotropy ($K_u \sim 1.3 \times 10^7$ erg cm^{-3}) [1] and high Curie temperature ($T_C \geq 823$ K) [2]
- Several attempts are being made to achieve a high degree of chemical order in this alloy but are hindered by the slow diffusion of atoms at relatively low order-disorder transition temperature
- In the present study, we aim to investigate the effect of deviations from a fully ordered structure on the magnetic properties by means of first-principles-based density functional theory (DFT) and Monte Carlo (MC) simulations



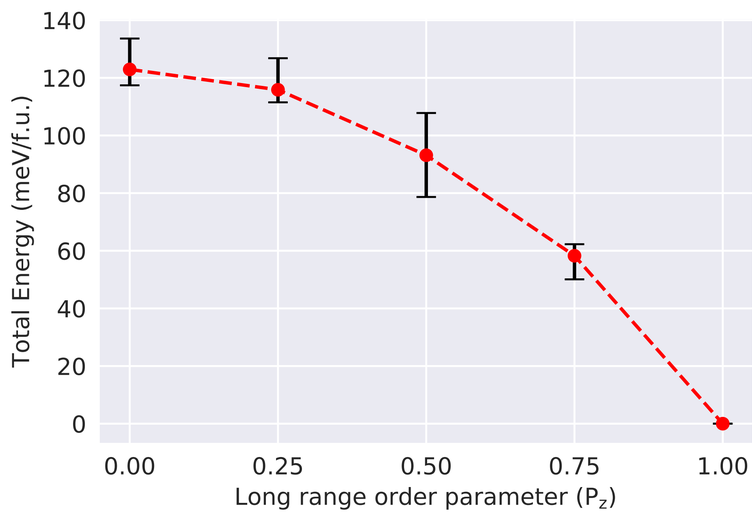
Iron meteorite (NWA 6259) $L1_0$ FeNi (Tetrataenite)

https://www.strahlen.org/tw/meteorite/meteorite_eisen5.html

Density functional theory calculations

- We study the effect of magnetic state on the chemical order-disorder transition temperature in $L1_0$ FeNi alloy based on Density functional theory (DFT) total energy calculations
- Degree of chemical order is taken into account by defining the long range order parameter P as $P^2 = P_x^2 + P_y^2 + P_z^2$, where P_x, P_y, and P_z are the long range order parameters along x, y, and z directions, respectively
- $P_z = 2$ * (probability that a Fe (Ni) atom sits on a Fe (Ni) sublattice) - 1. P_x and P_y are defined similarly
- $2 \times 2 \times 2$ supercells are constructed for different values of P_z with $P_x = P_y = 0$
- Total energies are calculated for different configurations for two magnetic states (a) ferromagnetic, and (b) paramagnetic.

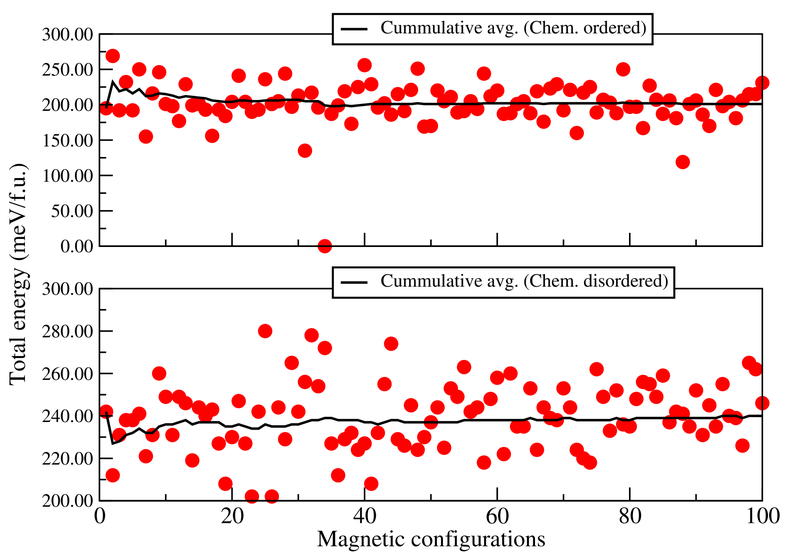
Ferromagnetic

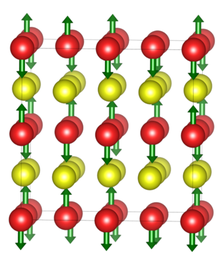


- At 0 K, fully ordered ferromagnetic FeNi phase is more stable than disordered and partially ordered phases

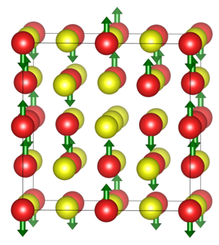
Paramagnetic

- To model the paramagnetic phase, we employed the so-called disordered local moment method [3] (DLM) using a supercell approach
- The energy of a paramagnetic phase is calculated by taking the average of the energies of a large set of DLM configurations for both ordered and disordered phases





Chem. ordered paramagnetic



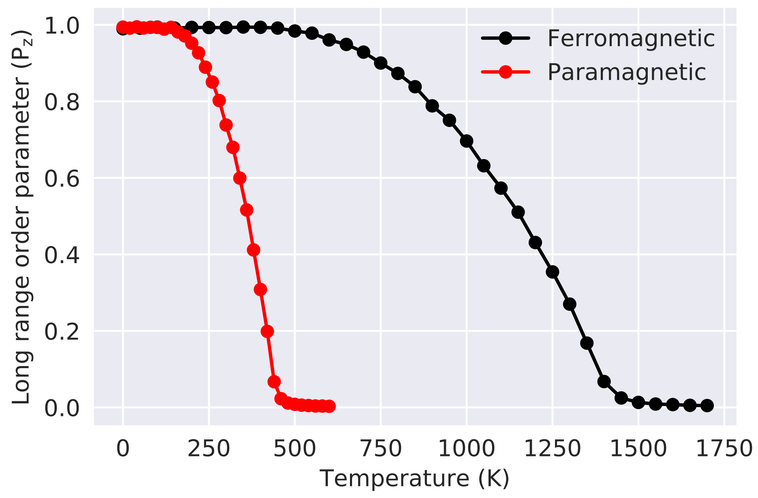
Chem. disordered paramagnetic

	$L1_0$ - FeNi (chemically ordered)	A1 FeNi (chemically disordered)
Ferromagnetic	0.00 meV	124 meV
	$T_C^{(ordered)}$ ↓ $E^{ordered}$	$T_C^{(disordered)}$ ↓ $E^{disordered}$
Paramagnetic	201 meV	240 meV

- $\Delta E^{ordered} > \Delta E^{disordered}$ $\Longrightarrow$ Curie temperature (ordered) > Curie temperature (disordered)

Monte Carlo simulations

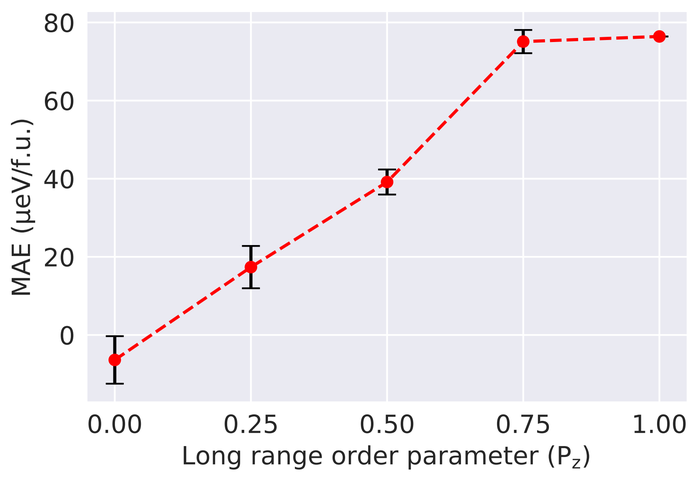
- To investigate the dependence of the chemical order-disorder transition temperature on the magnetic state, we employ Monte Carlo (MC) simulations
- $10 \times 10 \times 10$ supercell
- Total energy calculated by $E = -\Delta E P^2$, where $P^2 = P_x^2 + P_y^2 + P_z^2$ and ΔE is the energy difference between chemically-ordered and disordered phase



- Chemical order-disorder transition temperature (ferromagnetic) > Chemical order-disorder transition temperature (paramagnetic)
- Strong effect of magnetic state on order-disorder transition temperature

Magneto-Crystalline Anisotropy

- Magneto-crystalline anisotropy energy (MAE) is the difference in energy corresponding to different orientations of the magnetization relative to the crystallographic axes
- High MAE is one of the critical requirements for a permanent magnet
- Aim: To investigate the dependence of MAE on chemical ordering



- $MAE = E_{[100]} - E_{[001]}$, where $E_{[100]}$ and $E_{[001]}$ are the energies when magnetization is aligned along x and z axis, respectively
- No significant drop in MAE for $P_z = 0.75$!
- For $P_z = 0.75$, some of the configurations have higher MAE than the ordered structure (76 μeV/f.u) $\longrightarrow$ possible to achieve higher MAE ?

Summary and Conclusions

- We employed density-functional theory calculations in combination with Monte Carlo (MC) simulations to investigate the interplay between chemical order and the magnetic properties of the $L1_0$ FeNi phase.
- Our calculations for fixed ferromagnetic and paramagnetic configurations demonstrate a strong effect of the magnetic order on the chemical order-disorder transition temperature.
- Conversely, our results indicate that the chemically-ordered phase has a higher Curie temperature than the chemically-disordered phase.
- No significant drop in MAE for $P_z = 0.75$!

References

1. Kojima, T.*et al.*, *J. Phys.Condens.Matter* **26**, 064207 (2014)
2. Lewis, L.H.*et al.*, *J. Phys.Condens.Matter* **26**, 064213 (2014)
3. Gyorffy, B.L.*et al.*, *J. Phys.F: Met Phys.* **15**, 1337 (1985)